Learning Flowcharts

A **flowchart** is a picture of an algorithm: the steps to solve a problem, drawn as shapes joined by arrows. It lets you plan the logic before writing any code.

The five basic symbols

Shape	Meaning
Oval	Start or Stop. Every flowchart has exactly one of each.
Parallelogram	Input or Output (read a number, print a result).
Rectangle	Process: a calculation or assignment, e.g. `sum = sum + 1`.
Diamond	Decision: a Yes/No question. One arrow in, two arrows out.
Arrow	Shows the direction of flow.

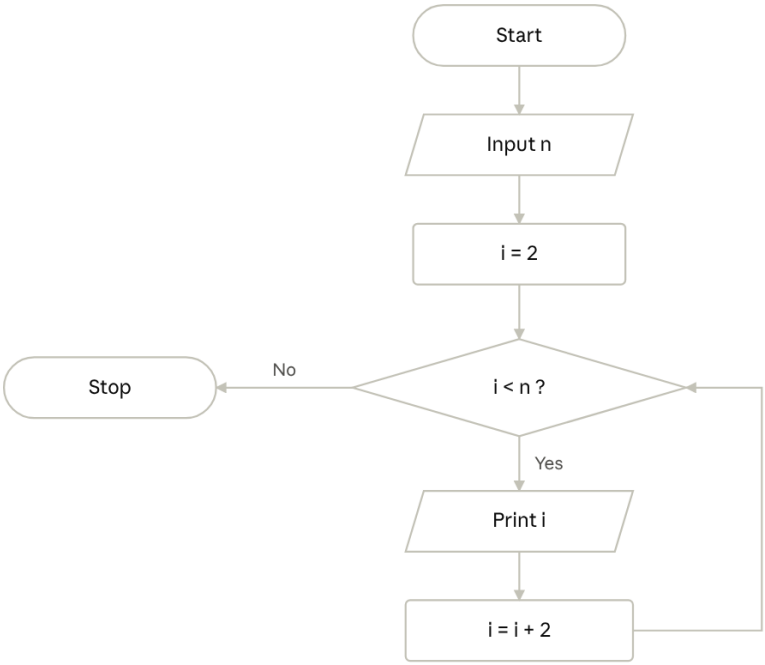
Rules to remember

- Flow runs top to bottom (or left to right).
- Begin with Start, finish with Stop.
- Each box holds one clear action.
- Decisions are labelled **Yes** and **No**.
- **Loops** are made by an arrow that goes back up to a decision. Most examples below use one.
- `mod` means remainder (7 mod 3 = 1). `div` means whole-number division (47 div 10 = 4).

1. Display all even numbers less than n

Start at 2 and add 2 each time while the number is still below n.

Even numbers: add 2 each pass until i reaches n



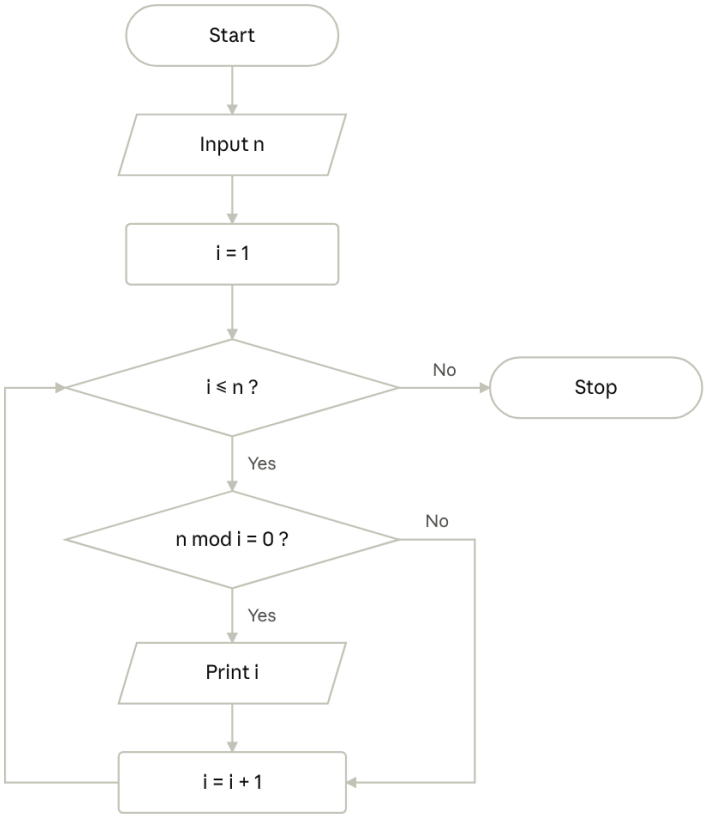
flowchart · even numbers below n, one loop

Trace: n = 9 prints 2, 4, 6, 8.

2. Find all factors of a number

Test every i from 1 to n. If n divides evenly by i, it is a factor.

Factors: test every i from 1 to n and print the ones that divide evenly



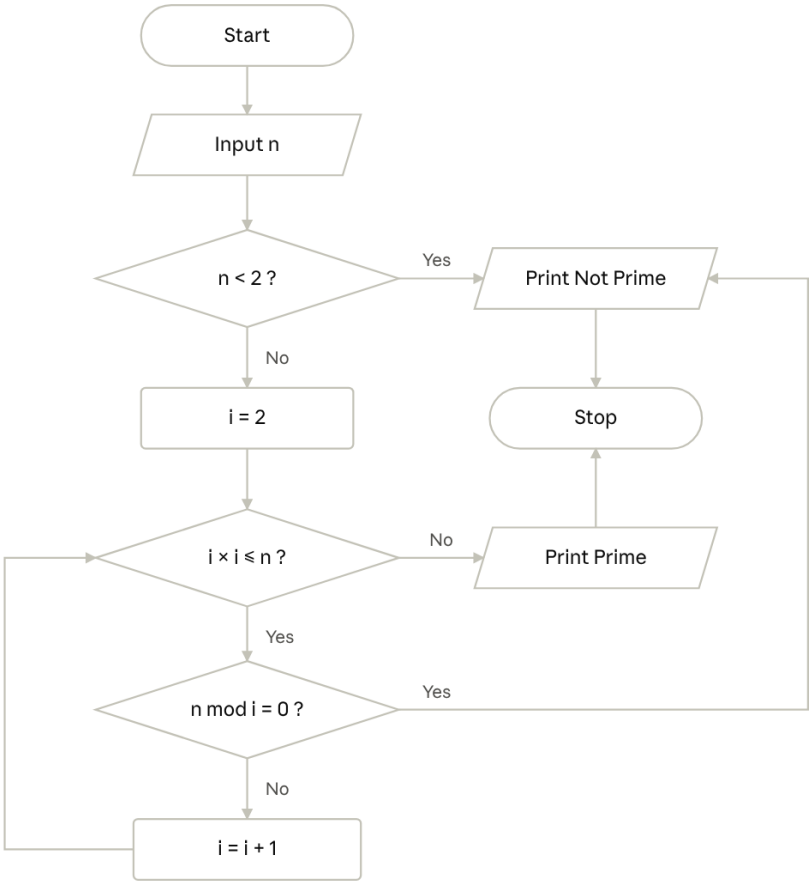
flowchart · factors of n, two decisions, one loop

Trace: n = 12 prints 1, 2, 3, 4, 6, 12.

3. Check if a number is prime

A prime has no divisor other than 1 and itself. Try divisors from 2 while i × i ≤ n; one hit means not prime.

Prime: a single divisor found means Not Prime



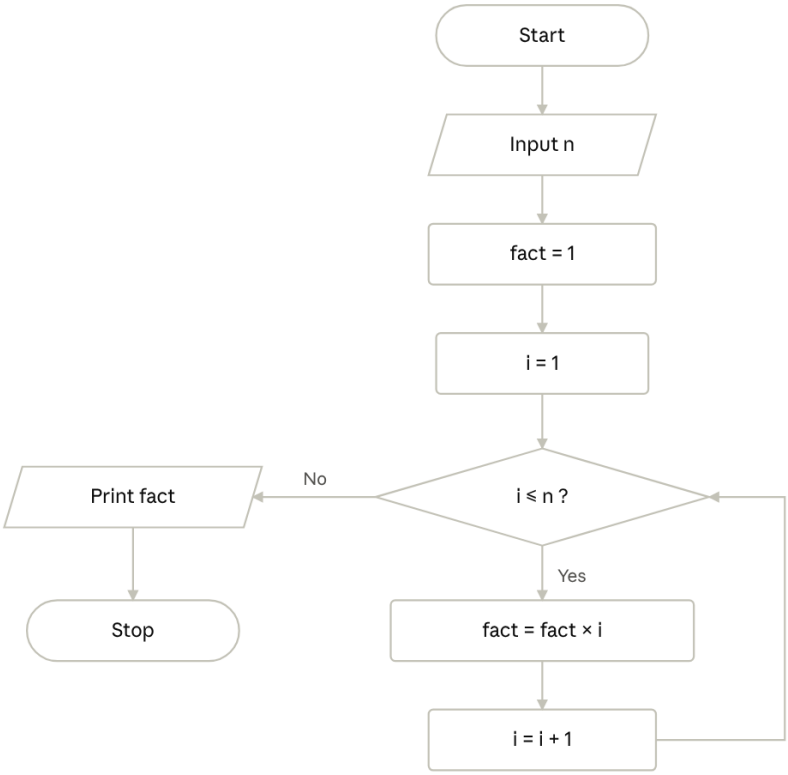
flowchart · prime check, three decisions, one loop

Trace: n = 7 → i = 2 (no), i = 3 stops the loop (9 > 7) → Prime.

4. Factorial of a number

n! = 1 × 2 × 3 × … × n. Keep a running product.

Factorial: multiply a running product by i, from 1 up to n



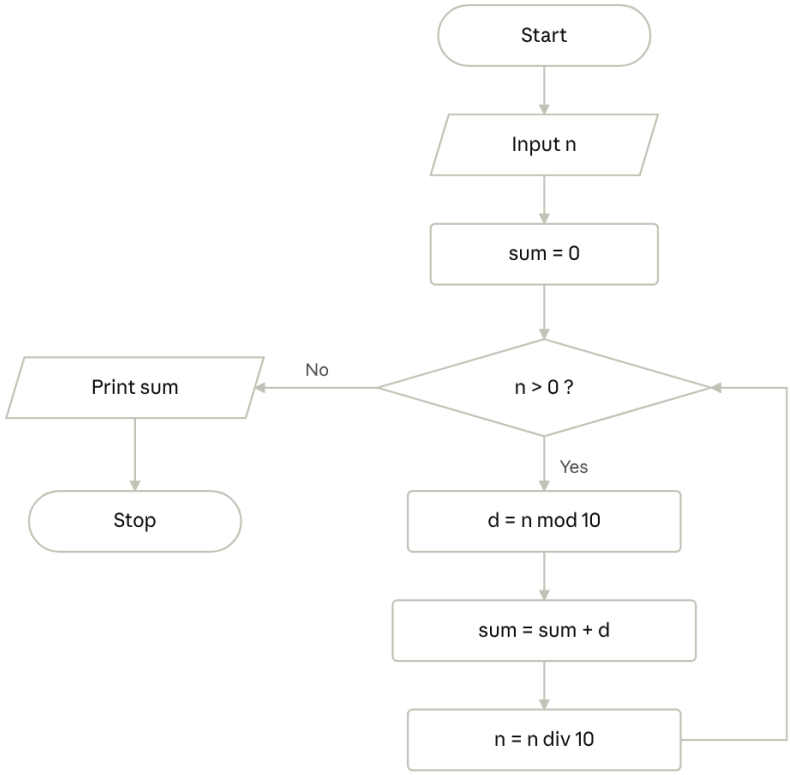
flowchart · factorial, one loop

Trace: n = 5 gives 1, 2, 6, 24, 120.

5. Sum of the digits of a number

Peel off the last digit with `mod 10`, add it, then drop it with `div 10`.

Digit sum: peel off the last digit until nothing is left



flowchart · digit sum, one loop

Trace: n = 482 → digits 2, 8, 4 → sum = 14.